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CREATE DATABASE company_db;
USE company_db;
CREATE TABLE employees (
emp_id INT PRIMARY KEY,
emp_name VARCHAR(50),
department VARCHAR(50),
salary INT,
hire_date DATE
);
INSERT INTO employees
(emp_id, emp_name, department, salary, hire_date)
VALUES
(1, 'Arun', 'IT', 90000, '2024-01-10'),
(2, 'Bala', 'IT', 80000, '2024-02-15'),
(3, 'Charan', 'IT', 80000, '2024-03-20'),
(4, 'Divya', 'HR', 75000, '2024-01-12'),
(5, 'Esha', 'HR', 70000, '2024-04-01'),
(6, 'Farhan', 'HR', 70000, '2024-05-05'),
(7, 'Gokul', 'Sales', 95000, '2024-02-02'),
(8, 'Hari', 'Sales', 85000, '2024-06-18');
SELECT * FROM employees;
-- 1.
SELECT ROW_NUMBER() OVER ( ORDER BY salary DESC ) AS row_num ,emp_id,
emp_name, department, salary FROM employees;
-- 2.
SELECT RANK() OVER ( ORDER BY salary DESC ) AS row_num ,emp_id, emp_name,
department, salary FROM employees;
-- 3.
SELECT DENSE_RANK() OVER ( ORDER BY salary DESC ) AS row_num ,emp_id,
emp_name, department, salary FROM employees;
-- 4.SELECT ROW_NUMBER() OVER ( PARTITION BY department ORDER BY salary
DESC ) AS
row_num, emp_name, department, salary FROM employees;
-- 5.
SELECT RANK() OVER ( PARTITION BY department ORDER BY salary DESC ) AS
salary_rank, emp_name, department, salary FROM employees;
-- 6.
SELECT * FROM (
SELECT emp_id, emp_name, department, salary, DENSE_RANK() OVER ( PARTITION BY
department ORDER BY salary DESC ) AS salary_rank FROM employees
) AS x
WHERE salary_rank <= 2;
-- 7.
SELECT emp_name, department, salary, FIRST_VALUE(emp_name) OVER ( PARTITION
BY
department ORDER BY salary DESC ) AS highest_paid_employee FROM employees;

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-- 8.

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SELECT emp_name, department, salary, LAST_VALUE(emp_name) OVER ( PARTITION  
BY  
department ORDER BY salary DESC ROWS BETWEEN UNBOUNDED PRECEDING AND  
UNBOUNDED FOLLOWING ) AS lowest_paid_employee  
FROM employees;
```

-- 9.

```
SELECT emp_name, department, salary,  
FIRST_VALUE(salary) OVER ( PARTITION BY department ORDER BY salary DESC ) AS  
highest_salary,  
LAST_VALUE(salary) OVER ( PARTITION BY department ORDER BY salary DESC ROWS  
BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING ) AS  
lowest_salary  
FROM employees;
```

-- 10.

```
SELECT * FROM (  
SELECT emp_id, emp_name, department, salary,  
DENSE_RANK() OVER ( ORDER BY salary DESC ) AS salary_rank FROM employees )  
AS x  
WHERE salary_rank = 2;
```